Appendix: R output (QCA package, fsQCA of 88 Australian SA4 regions)

Output of the Chapter 3 (Essay 2) analysis script: descriptive statistics, truth tables, and parsimonious (include = "?") and intermediate solutions for Homeostasis and High-Growth Entrepreneurship. Parsimonious solutions list all 40 tied models (M01-M40). Intermediate solutions are listed per group of parsimonious models (C1P). Method: CCubes, QCA 3.25.

```
Loading required package: admisc

To cite package QCA in publications, please use:
  Dusa, Adrian (2019) QCA with R. A Comprehensive Resource.
  Springer International Publishing.

Loaded 88 cases, 9 causal conditions, 2 outcomes.

Descriptive statistics of the transformed variables (Table 3.2):
                         N  Mean    SD   Min   P25 Median   P75   Max
Intermediaries              88 0.935 1.266 0.122 0.262  0.392 0.806 5.000
Finance                     88 0.916 1.087 0.132 0.264  0.388 0.980 5.000
Digital.infrastructure      88 1.000 0.932 0.075 0.310  0.639 1.440 4.171
Physical.infrastructure     88 1.000 1.035 0.093 0.341  0.511 1.368 4.807
Talent                      88 0.986 1.233 0.109 0.269  0.473 0.794 5.000
Demand                      88 0.693 0.985 0.089 0.202  0.360 0.730 5.000
Culture                     88 0.719 0.982 0.050 0.256  0.445 0.707 5.000
New.knowledge               88 1.000 0.631 0.055 0.436  1.400 1.697 1.697
Networks                    88 0.965 1.227 0.250 0.252  0.349 0.663 5.000
Homeostasis                 88 1.000 0.182 0.721 0.882  0.964 1.061 1.667
Highgrowth.entrepreneurship 88 0.716 1.083 0.027 0.199  0.361 0.646 5.000

Correlation matrix of the transformed variables (Table 3.3):
                         Homeostasis Highgrowth.entrepreneurship
Homeostasis                     1.00                       0.78
Highgrowth.entrepreneurship     0.78                       1.00
Intermediaries                  0.56                       0.68
Finance                         0.41                       0.45
Digital.infrastructure          0.47                       0.50
Physical.infrastructure         0.45                       0.53
Talent                          0.68                       0.81
Demand                          0.61                       0.75
Culture                         0.18                      -0.03
New.knowledge                   0.21                       0.08
Networks                        0.13                      -0.01
                         Intermediaries Finance Digital.infrastructure
Homeostasis                        0.56    0.41                   0.47
Highgrowth.entrepreneurship        0.68    0.45                   0.50
Intermediaries                     1.00    0.71                   0.70
Finance                            0.71    1.00                   0.38
Digital.infrastructure             0.70    0.38                   1.00
Physical.infrastructure            0.66    0.77                   0.44
Talent                             0.90    0.51                   0.78
Demand                             0.55    0.45                   0.41
Culture                           -0.06    0.14                   0.02
New.knowledge                      0.34    0.61                   0.06
Networks                          -0.08   -0.12                   0.02
                         Physical.infrastructure Talent Demand Culture
Homeostasis                                 0.45   0.68   0.61    0.18
Highgrowth.entrepreneurship                 0.53   0.81   0.75   -0.03
Intermediaries                              0.66   0.90   0.55   -0.06
Finance                                     0.77   0.51   0.45    0.14
Digital.infrastructure                      0.44   0.78   0.41    0.02
Physical.infrastructure                     1.00   0.57   0.47    0.14
Talent                                      0.57   1.00   0.60   -0.05
Demand                                      0.47   0.60   1.00    0.13
```

```
Culture                                    0.14  -0.05   0.13    1.00
New.knowledge                              0.31   0.17   0.03    0.27
Networks                                  -0.05   0.02   0.10    0.54
                          New.knowledge Networks
Homeostasis                        0.21     0.13
Highgrowth.entrepreneurship        0.08    -0.01
Intermediaries                     0.34    -0.08
Finance                            0.61    -0.12
Digital.infrastructure             0.06     0.02
Physical.infrastructure            0.31    -0.05
Talent                             0.17     0.02
Demand                             0.03     0.10
Culture                            0.27     0.54
New.knowledge                      1.00     0.07
Networks                           0.07     1.00


=======================================
Homeostasis (main analysis) | incl.cut = 0.8 | pri.cut = 0.65 | n.cut = 3
=======================================


  OUT: output value
    n: number of cases in configuration
 incl: sufficiency inclusion score
  PRI: proportional reduction in inconsistency

     Intermediaries Finance Digital.infrastructure Physical.infrastructure
512               1       1                     1                        1
511               1       1                     1                        1
506               1       1                     1                        1
378               1       0                     1                        1
507               1       1                     1                        1
135               0       1                     0                        0
  8               0       0                     0                        0
  2               0       0                     0                        0
 10               0       0                     0                        0
  1               0       0                     0                        0
     Talent Demand Culture New.knowledge Networks   OUT    n  incl  PRI
512     1      1       1           1          1        1     8  0.980 0.973
511     1      1       1           1          0        1     4  0.913 0.848
506     1      1       0           0          1        1     3  0.872 0.778
378     1      1       0           0          1        1     5  0.815 0.735
507     1      1       0           1          0        0     6  0.771 0.673
135     0      0       1           1          0        0     8  0.588 0.381
  8     0      0       1           1          1        0     4  0.572 0.329
  2     0      0       0           0          1        0     8  0.435 0.197
 10     0      1       0           0          1        0     3  0.418 0.071
  1     0      0       0           0          0        0     3  0.363 0.162
     cases
512  Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne -
West
511  Sydney - Blacktown,Sydney - City and Inner South,Sydney - Inner South West,Sydney - Parramatta
506  Brisbane - South,Brisbane Inner City,Perth - North West
378  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West
507  Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
135  Capital Region,Central West,Coffs Harbour - Grafton,Far West and Orana,Hunter Valley exc Newcastle,Murray,New England and North West,Riverina
  8  Hume,Latrobe - Gippsland,Shepparton,Warrnambool and South West
  2  Cairns,Darling Downs - Maranoa,Queensland - Outback,Toowoomba,Townsville,West and North West,Western Australia - Wheat Belt,Wide Bay
 10  Mackay - Isaac - Whitsunday,Western Australia - Outback (North),Western Australia - Outback (South)
  1  Barossa - Yorke - Mid North,South Australia - Outback,South Australia - South East


Configurations meeting n.cut: 10; treated as remainders: 502 of 512 (98%)
```

--- COMPLEX SOLUTION ---

M1: Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*Culture*New.knowledge +
    Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*~New.knowledge*Networks
    -> Homeostasis

		inclS	PRI	covS	covU
1	Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*Culture*New.knowledge	0.948	0.928	0.242	0.206
2	Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*~New.knowledge*Networks	0.829	0.752	0.137	0.101
	M1	0.899	0.862	0.344	

		cases
1	Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*Culture*New.knowledge	Sydney - Blacktown,Sydney - City and Inner South,Sydney - Inner South West,Sydney - Parramatta; Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West
2	Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*~New.knowledge*Networks	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West

--- PARSIMONIOUS SOLUTION ---

M01: Intermediaries*Culture + Intermediaries*~New.knowledge -> Homeostasis
M02: Intermediaries*Culture + Intermediaries*Networks -> Homeostasis
M03: Intermediaries*Culture + Digital.infrastructure*~New.knowledge
     -> Homeostasis
M04: Intermediaries*Culture + Digital.infrastructure*Networks -> Homeostasis
M05: Intermediaries*Culture + Physical.infrastructure*~New.knowledge
     -> Homeostasis
M06: Intermediaries*Culture + Physical.infrastructure*Networks -> Homeostasis
M07: Intermediaries*Culture + Talent*~New.knowledge -> Homeostasis
M08: Intermediaries*Culture + Talent*Networks -> Homeostasis
M09: Intermediaries*~New.knowledge + Digital.infrastructure*Culture
     -> Homeostasis
M10: Intermediaries*~New.knowledge + Physical.infrastructure*Culture
     -> Homeostasis
M11: Intermediaries*~New.knowledge + Talent*Culture -> Homeostasis
M12: Intermediaries*~New.knowledge + Demand*Culture -> Homeostasis
M13: Intermediaries*Networks + Digital.infrastructure*Culture -> Homeostasis
M14: Intermediaries*Networks + Physical.infrastructure*Culture -> Homeostasis
M15: Intermediaries*Networks + Talent*Culture -> Homeostasis
M16: Intermediaries*Networks + Demand*Culture -> Homeostasis
M17: Digital.infrastructure*Culture + Digital.infrastructure*~New.knowledge
     -> Homeostasis
M18: Digital.infrastructure*Culture + Digital.infrastructure*Networks
     -> Homeostasis
M19: Digital.infrastructure*Culture + Physical.infrastructure*~New.knowledge
     -> Homeostasis
M20: Digital.infrastructure*Culture + Physical.infrastructure*Networks
     -> Homeostasis
M21: Digital.infrastructure*Culture + Talent*~New.knowledge -> Homeostasis
M22: Digital.infrastructure*Culture + Talent*Networks -> Homeostasis
M23: Digital.infrastructure*~New.knowledge + Physical.infrastructure*Culture
     -> Homeostasis
M24: Digital.infrastructure*~New.knowledge + Talent*Culture -> Homeostasis
M25: Digital.infrastructure*~New.knowledge + Demand*Culture -> Homeostasis
M26: Digital.infrastructure*Networks + Physical.infrastructure*Culture
     -> Homeostasis

M27: Digital.infrastructure*Networks + Talent*Culture -> Homeostasis
M28: Digital.infrastructure*Networks + Demand*Culture -> Homeostasis
M29: Physical.infrastructure*Culture + Physical.infrastructure*~New.knowledge
-> Homeostasis
M30: Physical.infrastructure*Culture + Physical.infrastructure*Networks
-> Homeostasis
M31: Physical.infrastructure*Culture + Talent*~New.knowledge -> Homeostasis
M32: Physical.infrastructure*Culture + Talent*Networks -> Homeostasis
M33: Physical.infrastructure*~New.knowledge + Talent*Culture -> Homeostasis
M34: Physical.infrastructure*~New.knowledge + Demand*Culture -> Homeostasis
M35: Physical.infrastructure*Networks + Talent*Culture -> Homeostasis
M36: Physical.infrastructure*Networks + Demand*Culture -> Homeostasis
M37: Talent*Culture + Talent*~New.knowledge -> Homeostasis
M38: Talent*Culture + Talent*Networks -> Homeostasis
M39: Talent*~New.knowledge + Demand*Culture -> Homeostasis
M40: Talent*Networks + Demand*Culture -> Homeostasis

						(M1)	(M2)
		inclS	PRI	covS	covU		
1	Intermediaries*Culture	0.853	0.812	0.437	0.006	0.221	0.156
2	Intermediaries*~New.knowledge	0.835	0.799	0.424	0.005	0.209	
3	Intermediaries*Networks	0.873	0.844	0.403	0.000		0.122
4	Digital.infrastructure*Culture	0.849	0.807	0.391	0.000		
5	Digital.infrastructure*~New.knowledge	0.714	0.663	0.430	0.002		
6	Digital.infrastructure*Networks	0.743	0.693	0.402	0.000		
7	Physical.infrastructure*Culture	0.777	0.725	0.385	0.001		
8	Physical.infrastructure*~New.knowledge	0.694	0.627	0.379	0.000		
9	Physical.infrastructure*Networks	0.759	0.707	0.396	0.000		
10	Talent*Culture	0.906	0.878	0.463	0.006		
11	Talent*~New.knowledge	0.795	0.752	0.455	0.002		
12	Talent*Networks	0.849	0.814	0.440	0.000		
13	Demand*Culture	0.881	0.849	0.414	0.004		
	M1	0.822	0.782	0.645			
	M2	0.830	0.784	0.559			
	M3	0.739	0.687	0.672			
	M4	0.742	0.682	0.587			
	M5	0.732	0.669	0.635			
	M6	0.753	0.689	0.586			
	M7	0.789	0.742	0.683			
	M8	0.807	0.757	0.604			
	M9	0.812	0.771	0.620			
	M10	0.773	0.726	0.630			
	M11	0.847	0.811	0.679			
	M12	0.831	0.794	0.648			
	M13	0.821	0.775	0.543			
	M14	0.772	0.718	0.542			
	M15	0.859	0.821	0.594			
	M16	0.840	0.799	0.562			
	M17	0.740	0.691	0.618			
	M18	0.741	0.682	0.523			
	M19	0.726	0.665	0.597			
	M20	0.749	0.685	0.548			
	M21	0.782	0.736	0.651			
	M22	0.801	0.750	0.569			
	M23	0.703	0.650	0.643			
	M24	0.765	0.716	0.698			
	M25	0.749	0.702	0.648			
	M26	0.697	0.634	0.547			
	M27	0.769	0.713	0.603			
	M28	0.754	0.698	0.559			
	M29	0.692	0.630	0.573			

M30	0.710	0.642	0.507
M31	0.746	0.695	0.657
M32	0.756	0.699	0.572
M33	0.758	0.699	0.658
M34	0.745	0.689	0.621
M35	0.782	0.723	0.606
M36	0.771	0.713	0.567
M37	0.822	0.781	0.687
M38	0.846	0.802	0.591
M39	0.801	0.759	0.678
M40	0.822	0.777	0.595

	(M3)	(M4)	(M5)	(M6)	(M7)	(M8)
Intermediaries*Culture	0.242	0.186	0.257	0.190	0.228	0.164
Intermediaries*~New.knowledge						
Intermediaries*Networks						
Digital.infrastructure*Culture						
Digital.infrastructure*~New.knowledge	0.235					
Digital.infrastructure*Networks		0.151				
Physical.infrastructure*Culture						
Physical.infrastructure*~New.knowledge			0.199			
Physical.infrastructure*Networks				0.150		
Talent*Culture						
Talent*~New.knowledge					0.247	
Talent*Networks						0.168
Demand*Culture						

	(M9)	(M10)	(M11)	(M12)	(M13)	(M14)
Intermediaries*Culture						
Intermediaries*~New.knowledge	0.230	0.244	0.216	0.234		
Intermediaries*Networks					0.152	0.157
Digital.infrastructure*Culture	0.196				0.140	
Digital.infrastructure*~New.knowledge						
Digital.infrastructure*Networks						
Physical.infrastructure*Culture		0.205				0.139
Physical.infrastructure*~New.knowledge						
Physical.infrastructure*Networks						
Talent*Culture			0.255			
Talent*~New.knowledge						
Talent*Networks						
Demand*Culture				0.224		

	(M15)	(M16)	(M17)	(M18)	(M19)	(M20)
Intermediaries*Culture						
Intermediaries*~New.knowledge						
Intermediaries*Networks	0.131	0.148				
Digital.infrastructure*Culture			0.188	0.121	0.218	0.151
Digital.infrastructure*~New.knowledge			0.227			
Digital.infrastructure*Networks				0.132		
Physical.infrastructure*Culture						
Physical.infrastructure*~New.knowledge					0.206	
Physical.infrastructure*Networks						0.157
Talent*Culture	0.191					

11	Talent*~New.knowledge	
12	Talent*Networks	
13	Demand*Culture	0.159

		(M21)	(M22)	(M23)	(M24)	(M25)	(M26)
1	Intermediaries*Culture						
2	Intermediaries*~New.knowledge						
3	Intermediaries*Networks						
4	Digital.infrastructure*Culture	0.196	0.129				
5	Digital.infrastructure*~New.knowledge			0.257	0.235	0.234	
6	Digital.infrastructure*Networks						0.162
7	Physical.infrastructure*Culture			0.213			0.146
8	Physical.infrastructure*~New.knowledge						
9	Physical.infrastructure*Networks						
10	Talent*Culture				0.268		
11	Talent*~New.knowledge	0.260					
12	Talent*Networks		0.178				
13	Demand*Culture					0.218	

		(M27)	(M28)	(M29)	(M30)	(M31)	(M32)
1	Intermediaries*Culture						
2	Intermediaries*~New.knowledge						
3	Intermediaries*Networks						
4	Digital.infrastructure*Culture						
5	Digital.infrastructure*~New.knowledge						
6	Digital.infrastructure*Networks	0.140	0.146				
7	Physical.infrastructure*Culture			0.195	0.110	0.202	0.132
8	Physical.infrastructure*~New.knowledge			0.188			
9	Physical.infrastructure*Networks				0.121		
10	Talent*Culture	0.201					
11	Talent*~New.knowledge					0.272	
12	Talent*Networks						0.187
13	Demand*Culture		0.158				

		(M33)	(M34)	(M35)	(M36)	(M37)	(M38)
1	Intermediaries*Culture						
2	Intermediaries*~New.knowledge						
3	Intermediaries*Networks						
4	Digital.infrastructure*Culture						
5	Digital.infrastructure*~New.knowledge						
6	Digital.infrastructure*Networks						
7	Physical.infrastructure*Culture						
8	Physical.infrastructure*~New.knowledge	0.195	0.207				
9	Physical.infrastructure*Networks			0.143	0.153		
10	Talent*Culture	0.279		0.210		0.232	0.151
11	Talent*~New.knowledge					0.224	
12	Talent*Networks						0.128
13	Demand*Culture		0.242		0.171		

		(M39)	(M40)

```
-------------------------------------------------------
 1                 Intermediaries*Culture
 2           Intermediaries*~New.knowledge
 3                 Intermediaries*Networks
 4          Digital.infrastructure*Culture
 5   Digital.infrastructure*~New.knowledge
 6         Digital.infrastructure*Networks
 7         Physical.infrastructure*Culture
 8  Physical.infrastructure*~New.knowledge
 9        Physical.infrastructure*Networks
10                          Talent*Culture
11                  Talent*~New.knowledge  0.264
12                        Talent*Networks         0.181
13                         Demand*Culture  0.223  0.155
-------------------------------------------------------

                                          cases
-------------------------------------------------
 1                 Intermediaries*Culture  Sydney - Blacktown,Sydney - City and Inner South,Sydney - Inner South West,Sydney - Parramatta;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
 2           Intermediaries*~New.knowledge  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West;
                                           Brisbane - South,Brisbane Inner City,Perth - North West
 3                 Intermediaries*Networks  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West;
                                           Brisbane - South,Brisbane Inner City,Perth - North West;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
 4          Digital.infrastructure*Culture  Sydney - Blacktown,Sydney - City and Inner South,Sydney - Inner South West,Sydney - Parramatta;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
 5   Digital.infrastructure*~New.knowledge  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West;
                                           Brisbane - South,Brisbane Inner City,Perth - North West
 6         Digital.infrastructure*Networks  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West;
                                           Brisbane - South,Brisbane Inner City,Perth - North West;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
 7         Physical.infrastructure*Culture  Sydney - Blacktown,Sydney - City and Inner South,Sydney - Inner South West,Sydney - Parramatta;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
 8  Physical.infrastructure*~New.knowledge  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West;
                                           Brisbane - South,Brisbane Inner City,Perth - North West
 9        Physical.infrastructure*Networks  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West;
                                           Brisbane - South,Brisbane Inner City,Perth - North West;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
10                          Talent*Culture  Sydney - Blacktown,Sydney - City and Inner South,Sydney - Inner South West,Sydney - Parramatta;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
11                  Talent*~New.knowledge  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West;
                                           Brisbane - South,Brisbane Inner City,Perth - North West
12                        Talent*Networks  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West;
                                           Brisbane - South,Brisbane Inner City,Perth - North West;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
13                         Demand*Culture  Sydney - Blacktown,Sydney - City and Inner South,Sydney - Inner South West,Sydney - Parramatta;
                                           Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer
East,Melbourne - South East,Melbourne - West
-------------------------------------------------


--- INTERMEDIATE SOLUTION ---

From C1P1, C1P3, C1P5, C1P7, C1P9, C1P10, C1P11, C1P12, C1P17, C1P19, C1P21, C1P23, C1P24, C1P25, C1P29, C1P31, C1P33, C1P34, C1P37, C1P39:
```

```
M1:    Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~New.knowledge*Networks +
       Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*Culture*New.knowledge
       -> Homeostasis

                                                
```

```
506         1           1             1                       1
378         1           0             1                       1
511         1           1             1                       1
  8         0           0             0                       0
 10         0           0             0                       0
  2         0           0             0                       0
  1         0           0             0                       0
135         0           1             0                       0
     Talent Demand Culture New.knowledge Networks   OUT    n  incl  PRI
512     1      1       1          1          1        1     8  0.985 0.979
507     1      1       0          1          0        1     6  0.958 0.944
506     1      1       0          0          1        1     3  0.928 0.879
378     1      1       0          0          1        1     5  0.910 0.865
511     1      1       1          1          0        0     4  0.846 0.753
  8     0      0       1          1          1        0     4  0.544 0.128
 10     0      1       0          0          1        0     3  0.447 0.052
  2     0      0       0          0          1        0     8  0.388 0.081
  1     0      0       0          0          0        0     3  0.339 0.070
135     0      0       1          1          0        0     8  0.314 0.096
     cases
512  Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne -
West
507  Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
506  Brisbane - South,Brisbane Inner City,Perth - North West
378  Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West
511  Sydney - Blacktown,Sydney - City and Inner South,Sydney - Inner South West,Sydney - Parramatta
  8  Hume,Latrobe - Gippsland,Shepparton,Warrnambool and South West
 10  Mackay - Isaac - Whitsunday,Western Australia - Outback (North),Western Australia - Outback (South)
  2  Cairns,Darling Downs - Maranoa,Queensland - Outback,Toowoomba,Townsville,West and North West,Western Australia - Wheat Belt,Wide Bay
  1  Barossa - Yorke - Mid North,South Australia - Outback,South Australia - South East
135  Capital Region,Central West,Coffs Harbour - Grafton,Far West and Orana,Hunter Valley exc Newcastle,Murray,New England and North West,Riverina


Configurations meeting n.cut: 10; treated as remainders: 502 of 512 (98%)

--- COMPLEX SOLUTION ---

M1: Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*~New.knowledge*Networks +
    Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*New.knowledge*~Networks +
    Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*Culture*New.knowledge*Networks
    -> Highgrowth.entrepreneurship

                                                
```

```
M01: Intermediaries*~Culture + Intermediaries*Networks
     -> Highgrowth.entrepreneurship
M02: Intermediaries*~Culture + Finance*Networks -> Highgrowth.entrepreneurship
M03: Intermediaries*~Culture + Digital.infrastructure*Networks
     -> Highgrowth.entrepreneurship
M04: Intermediaries*~Culture + Physical.infrastructure*Networks
     -> Highgrowth.entrepreneurship
M05: Intermediaries*~Culture + Talent*Networks -> Highgrowth.entrepreneurship
M06: Intermediaries*~Culture + Demand*Culture*Networks
     -> Highgrowth.entrepreneurship
M07: Intermediaries*~Culture + Demand*New.knowledge*Networks
     -> Highgrowth.entrepreneurship
M08: Intermediaries*Networks + Finance*~Culture -> Highgrowth.entrepreneurship
M09: Intermediaries*Networks + Digital.infrastructure*~Culture
     -> Highgrowth.entrepreneurship
M10: Intermediaries*Networks + Physical.infrastructure*~Culture
     -> Highgrowth.entrepreneurship
M11: Intermediaries*Networks + Talent*~Culture -> Highgrowth.entrepreneurship
M12: Intermediaries*Networks + ~Culture*New.knowledge
     -> Highgrowth.entrepreneurship
M13: Intermediaries*Networks + Demand*~Culture*~Networks
     -> Highgrowth.entrepreneurship
M14: Finance*~Culture + Digital.infrastructure*Networks
     -> Highgrowth.entrepreneurship
M15: Finance*~Culture + Physical.infrastructure*Networks
     -> Highgrowth.entrepreneurship
M16: Finance*~Culture + Talent*Networks -> Highgrowth.entrepreneurship
M17: Finance*Networks + Digital.infrastructure*~Culture
     -> Highgrowth.entrepreneurship
M18: Finance*Networks + Physical.infrastructure*~Culture
     -> Highgrowth.entrepreneurship
M19: Finance*Networks + Talent*~Culture -> Highgrowth.entrepreneurship
M20: Digital.infrastructure*~Culture + Digital.infrastructure*Networks
     -> Highgrowth.entrepreneurship
M21: Digital.infrastructure*~Culture + Physical.infrastructure*Networks
     -> Highgrowth.entrepreneurship
M22: Digital.infrastructure*~Culture + Talent*Networks
     -> Highgrowth.entrepreneurship
M23: Digital.infrastructure*~Culture + Demand*Culture*Networks
     -> Highgrowth.entrepreneurship
M24: Digital.infrastructure*~Culture + Demand*New.knowledge*Networks
     -> Highgrowth.entrepreneurship
M25: Digital.infrastructure*Networks + Physical.infrastructure*~Culture
     -> Highgrowth.entrepreneurship
M26: Digital.infrastructure*Networks + Talent*~Culture
     -> Highgrowth.entrepreneurship
M27: Digital.infrastructure*Networks + ~Culture*New.knowledge
     -> Highgrowth.entrepreneurship
M28: Digital.infrastructure*Networks + Demand*~Culture*~Networks
     -> Highgrowth.entrepreneurship
M29: Physical.infrastructure*~Culture + Physical.infrastructure*Networks
     -> Highgrowth.entrepreneurship
M30: Physical.infrastructure*~Culture + Talent*Networks
     -> Highgrowth.entrepreneurship
M31: Physical.infrastructure*~Culture + Demand*Culture*Networks
     -> Highgrowth.entrepreneurship
M32: Physical.infrastructure*~Culture + Demand*New.knowledge*Networks
     -> Highgrowth.entrepreneurship
M33: Physical.infrastructure*Networks + Talent*~Culture
     -> Highgrowth.entrepreneurship
M34: Physical.infrastructure*Networks + ~Culture*New.knowledge
     -> Highgrowth.entrepreneurship
M35: Physical.infrastructure*Networks + Demand*~Culture*~Networks
```

```
     -> Highgrowth.entrepreneurship
M36: Talent*~Culture + Talent*Networks -> Highgrowth.entrepreneurship
M37: Talent*~Culture + Demand*Culture*Networks -> Highgrowth.entrepreneurship
M38: Talent*~Culture + Demand*New.knowledge*Networks
     -> Highgrowth.entrepreneurship
M39: Talent*Networks + ~Culture*New.knowledge -> Highgrowth.entrepreneurship
M40: Talent*Networks + Demand*~Culture*~Networks -> Highgrowth.entrepreneurship
```

					----------	----------	
		inclS	PRI	covS	covU	(M1)	(M2)
1	Intermediaries*~Culture	0.832	0.793	0.428	0.000	0.223	0.296
2	Intermediaries*Networks	0.883	0.846	0.435	0.003	0.230	
3	Finance*~Culture	0.881	0.837	0.310	0.000		
4	Finance*Networks	0.915	0.883	0.353	0.001		0.220
5	Digital.infrastructure*~Culture	0.746	0.686	0.458	0.000		
6	Digital.infrastructure*Networks	0.799	0.739	0.461	0.000		
7	Physical.infrastructure*~Culture	0.732	0.676	0.417	0.000		
8	Physical.infrastructure*Networks	0.766	0.709	0.427	0.005		
9	Talent*~Culture	0.827	0.790	0.465	0.001		
10	Talent*Networks	0.848	0.805	0.469	0.012		
11	~Culture*New.knowledge	0.848	0.783	0.210	0.000		
12	Demand*~Culture*~Networks	0.806	0.738	0.326	0.000		
13	Demand*Culture*Networks	0.908	0.881	0.279	0.000		
14	Demand*New.knowledge*Networks	0.933	0.894	0.187	0.000		
	M1	0.861	0.829	0.659			
	M2	0.858	0.825	0.649			
	M3	0.795	0.749	0.690			
	M4	0.781	0.737	0.683			
	M5	0.825	0.786	0.701			
	M6	0.856	0.826	0.650			
	M7	0.857	0.821	0.553			
	M8	0.875	0.839	0.613			
	M9	0.787	0.737	0.694			
	M10	0.787	0.738	0.680			
	M11	0.840	0.805	0.704			
	M12	0.871	0.835	0.580			
	M13	0.842	0.797	0.614			
	M14	0.811	0.760	0.635			
	M15	0.795	0.748	0.623			
	M16	0.849	0.808	0.649			
	M17	0.791	0.742	0.674			
	M18	0.791	0.745	0.656			
	M19	0.851	0.818	0.687			
	M20	0.791	0.746	0.682			
	M21	0.757	0.708	0.691			
	M22	0.770	0.719	0.718			
	M23	0.795	0.751	0.679			
	M24	0.785	0.731	0.583			
	M25	0.770	0.719	0.684			
	M26	0.792	0.747	0.716			
	M27	0.807	0.754	0.605			
	M28	0.799	0.748	0.625			
	M29	0.779	0.736	0.645			
	M30	0.779	0.731	0.704			
	M31	0.789	0.746	0.644			
	M32	0.778	0.725	0.546			
	M33	0.788	0.747	0.709			
	M34	0.789	0.739	0.575			
	M35	0.783	0.734	0.619			
	M36	0.840	0.807	0.714			
	M37	0.851	0.821	0.687			

```
                                M38  0.850  0.815  0.591
                                M39  0.843  0.802  0.614
                                M40  0.820  0.772  0.651


-------------------------------------------------------------------------------
                                    (M3)   (M4)   (M5)   (M6)   (M7)   (M8)
-------------------------------------------------------------------------------
 1           Intermediaries*~Culture  0.229  0.256  0.232  0.371  0.366
 2           Intermediaries*Networks                                      0.303
 3                 Finance*~Culture                                       0.178
 4                 Finance*Networks
 5   Digital.infrastructure*~Culture
 6   Digital.infrastructure*Networks  0.262
 7  Physical.infrastructure*~Culture
 8  Physical.infrastructure*Networks         0.254
 9                  Talent*~Culture
10                  Talent*Networks                0.273
11            ~Culture*New.knowledge
12         Demand*~Culture*~Networks
13           Demand*Culture*Networks                       0.222
14     Demand*New.knowledge*Networks                              0.125
-------------------------------------------------------------------------------


-------------------------------------------------------------------------------
                                    (M9)   (M10)  (M11)  (M12)  (M13)  (M14)
-------------------------------------------------------------------------------
 1           Intermediaries*~Culture
 2           Intermediaries*Networks  0.236  0.263  0.239  0.370  0.288
 3                 Finance*~Culture                                       0.174
 4                 Finance*Networks
 5   Digital.infrastructure*~Culture  0.259
 6   Digital.infrastructure*Networks                                      0.325
 7  Physical.infrastructure*~Culture         0.245
 8  Physical.infrastructure*Networks
 9                  Talent*~Culture                0.269
10                  Talent*Networks
11            ~Culture*New.knowledge                       0.145
12         Demand*~Culture*~Networks                              0.179
13           Demand*Culture*Networks
14     Demand*New.knowledge*Networks
-------------------------------------------------------------------------------


-------------------------------------------------------------------------------
                                   (M15)  (M16)  (M17)  (M18)  (M19)  (M20)
-------------------------------------------------------------------------------
 1           Intermediaries*~Culture
 2           Intermediaries*Networks
 3                 Finance*~Culture  0.196  0.179
 4                 Finance*Networks                0.216  0.239  0.222
 5   Digital.infrastructure*~Culture                0.322                0.221
 6   Digital.infrastructure*Networks                                     0.224
 7  Physical.infrastructure*~Culture                       0.303
 8  Physical.infrastructure*Networks  0.313
 9                  Talent*~Culture                              0.334
10                  Talent*Networks         0.338
11            ~Culture*New.knowledge
12         Demand*~Culture*~Networks
13           Demand*Culture*Networks
14     Demand*New.knowledge*Networks
-------------------------------------------------------------------------------
```

		(M21)	(M22)	(M23)	(M24)	(M25)	(M26)
1	Intermediaries*~Culture						
2	Intermediaries*Networks						
3	Finance*~Culture						
4	Finance*Networks						
5	Digital.infrastructure*~Culture	0.264	0.249	0.400	0.396		
6	Digital.infrastructure*Networks					0.267	0.251
7	Physical.infrastructure*~Culture					0.224	
8	Physical.infrastructure*Networks	0.233					
9	Talent*~Culture						0.255
10	Talent*Networks		0.260				
11	~Culture*New.knowledge						
12	Demand*~Culture*~Networks						
13	Demand*Culture*Networks			0.221			
14	Demand*New.knowledge*Networks				0.125		

		(M27)	(M28)	(M29)	(M30)	(M31)	(M32)
1	Intermediaries*~Culture						
2	Intermediaries*Networks						
3	Finance*~Culture						
4	Finance*Networks						
5	Digital.infrastructure*~Culture						
6	Digital.infrastructure*Networks	0.395	0.299				
7	Physical.infrastructure*~Culture			0.218	0.235	0.365	0.359
8	Physical.infrastructure*Networks			0.228			
9	Talent*~Culture						
10	Talent*Networks				0.287		
11	~Culture*New.knowledge	0.145					
12	Demand*~Culture*~Networks		0.164				
13	Demand*Culture*Networks					0.227	
14	Demand*New.knowledge*Networks						0.129

		(M33)	(M34)	(M35)	(M36)	(M37)	(M38)
1	Intermediaries*~Culture						
2	Intermediaries*Networks						
3	Finance*~Culture						
4	Finance*Networks						
5	Digital.infrastructure*~Culture						
6	Digital.infrastructure*Networks						
7	Physical.infrastructure*~Culture						
8	Physical.infrastructure*Networks	0.244	0.365	0.293			
9	Talent*~Culture	0.282			0.245	0.408	0.404
10	Talent*Networks				0.249		
11	~Culture*New.knowledge		0.148				
12	Demand*~Culture*~Networks			0.192			
13	Demand*Culture*Networks					0.222	
14	Demand*New.knowledge*Networks						0.126

		(M39)	(M40)
1	Intermediaries*~Culture		

2	Intermediaries*Networks		
3	Finance*~Culture		
4	Finance*Networks		
5	Digital.infrastructure*~Culture		
6	Digital.infrastructure*Networks		
7	Physical.infrastructure*~Culture		
8	Physical.infrastructure*Networks		
9	Talent*~Culture		
10	Talent*Networks	0.404	0.325
11	~Culture*New.knowledge	0.145	
12	Demand*~Culture*~Networks		0.182
13	Demand*Culture*Networks		
14	Demand*New.knowledge*Networks		

		cases
1	Intermediaries*~Culture	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
2	Intermediaries*Networks	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West
3	Finance*~Culture	Brisbane - South,Brisbane Inner City,Perth - North West; Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
4	Finance*Networks	Brisbane - South,Brisbane Inner City,Perth - North West; Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West
5	Digital.infrastructure*~Culture	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
6	Digital.infrastructure*Networks	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West
7	Physical.infrastructure*~Culture	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
8	Physical.infrastructure*Networks	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West
9	Talent*~Culture	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
10	Talent*Networks	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West
11	~Culture*New.knowledge	Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
12	Demand*~Culture*~Networks	Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
13	Demand*Culture*Networks	Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West
14	Demand*New.knowledge*Networks	Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West

--- INTERMEDIATE SOLUTION ---

From C1P1, C1P3, C1P4, C1P5, C1P8, C1P9, C1P10, C1P11, C1P12, C1P13, C1P14, C1P15, C1P16, C1P20, C1P21, C1P22, C1P25, C1P26, C1P27, C1P28, C1P29, C1P30, C1P33, C1P34, C1P35,

C1P36, C1P39, C1P40:

```
M1:    Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*Networks +
       Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*New.knowledge
       -> Highgrowth.entrepreneurship
```

		inclS	PRI	covS	covU
1	Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*Networks	0.939	0.921	0.347	0.297
2	Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*New.knowledge	0.959	0.945	0.180	0.130
	M1	0.947	0.934	0.476	

		cases
1	Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*Networks	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West; Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West
2	Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*New.knowledge	Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland

From C1P2, C1P6, C1P7, C1P17, C1P18, C1P19, C1P23, C1P24, C1P31, C1P32, C1P37, C1P38:

```
M1:    Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*Networks +
       Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*New.knowledge +
       Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*New.knowledge*Networks
       -> Highgrowth.entrepreneurship
```

		inclS	PRI	covS	covU
1	Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*Networks	0.920	0.885	0.163	0.113
2	Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*New.knowledge	0.959	0.945	0.180	0.130
3	Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*New.knowledge*Networks	0.978	0.967	0.162	0.112
	M1	0.957	0.945	0.405	

		cases
1	Intermediaries*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*Networks	Brisbane - North,Brisbane - West,Perth - Inner,Perth - South East,Perth - South West; Brisbane - South,Brisbane Inner City,Perth - North West
2	Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*~Culture*New.knowledge	Sydney - Eastern Suburbs,Sydney - Inner West,Sydney - North Sydney and Hornsby,Sydney - Northern Beaches,Sydney - Ryde,Sydney - Sutherland
3	Intermediaries*Finance*Digital.infrastructure*Physical.infrastructure*Talent*Demand*New.knowledge*Networks	Melbourne - Inner,Melbourne - Inner East,Melbourne - Inner South,Melbourne - North East,Melbourne - North West,Melbourne - Outer East,Melbourne - South East,Melbourne - West

--- NECESSITY: Homeostasis (presence, then absence) ---

		inclN	RoN	covN
1	Intermediaries	0.724	0.835	0.782
2	Finance	0.671	0.770	0.696
3	Digital.infrastructure	0.696	0.784	0.720
4	Physical.infrastructure	0.651	0.762	0.678
5	Talent	0.767	0.824	0.787
6	Demand	0.701	0.804	0.741
7	Culture	0.597	0.732	0.622
8	New.knowledge	0.508	0.777	0.613

```
9                 Networks  0.543  0.713  0.574
-----------------------------------------------


                            inclN   RoN   covN
-----------------------------------------------
1           ~Intermediaries  0.382  0.591  0.374
2                  ~Finance  0.403  0.624  0.409
3   ~Digital.infrastructure  0.388  0.619  0.395
4  ~Physical.infrastructure  0.445  0.638  0.450
5                   ~Talent  0.333  0.603  0.342
6                   ~Demand  0.404  0.612  0.403
7                  ~Culture  0.484  0.655  0.490
8            ~New.knowledge  0.559  0.597  0.499
9                 ~Networks  0.539  0.671  0.538
-----------------------------------------------


--- NECESSITY: High-growth entrepreneurship (presence, then absence) ---

                           inclN   RoN   covN
-----------------------------------------------
1           Intermediaries  0.801  0.854  0.811
2                  Finance  0.693  0.758  0.674
3   Digital.infrastructure  0.794  0.816  0.770
4  Physical.infrastructure  0.746  0.791  0.728
5                   Talent  0.856  0.850  0.824
6                   Demand  0.774  0.819  0.767
7                  Culture  0.546  0.689  0.534
8            New.knowledge  0.521  0.766  0.588
9                 Networks  0.555  0.702  0.549
-----------------------------------------------


                            inclN   RoN   covN
-----------------------------------------------
1           ~Intermediaries  0.320  0.561  0.293
2                  ~Finance  0.401  0.613  0.382
3   ~Digital.infrastructure  0.325  0.588  0.310
4  ~Physical.infrastructure  0.346  0.591  0.328
5                   ~Talent  0.249  0.568  0.239
6                   ~Demand  0.342  0.581  0.320
7                  ~Culture  0.523  0.658  0.495
8            ~New.knowledge  0.581  0.591  0.487
9                 ~Networks  0.556  0.662  0.519
-----------------------------------------------


Done. Files written to: /tmp/claude-0/-home-claude/3a116361-0765-580c-ba41-76d5670d33d1/scratchpad/rrun
```